

Chapter 18: Further Differentiability Results

Calculus of Subdifferentials in Convex Analysis

Based on Bauschke & Combettes (2017)

2024

Chapter Overview

- 1 18.1 The Ekeland-Lebourn Theorem
- 2 18.2 The Subdifferential of a Maximum
- 3 18.3 Differentiability of Infimal Convolutions
- 4 18.4 Differentiability and Strict Convexity
- 5 18.5 Stronger Notions of Differentiability
- 6 18.6 Differentiability of the Distance to a Set
- 7 Numerical Examples and Applications
- 8 Summary and Applications

The Ekeland-Lebourn Theorem I

Key Idea: For a convex function that is continuous at a point, Fréchet differentiability holds on a *dense* G_δ subset of the domain.

Theorem (Ekeland-Lebourn)

Let $f : \mathcal{H} \rightarrow [-\infty, +\infty]$ be convex, and suppose that $\text{cont } f \neq \emptyset$. Then the set of points at which f is **Fréchet differentiable** is a dense G_δ subset of $\text{dom } f$.

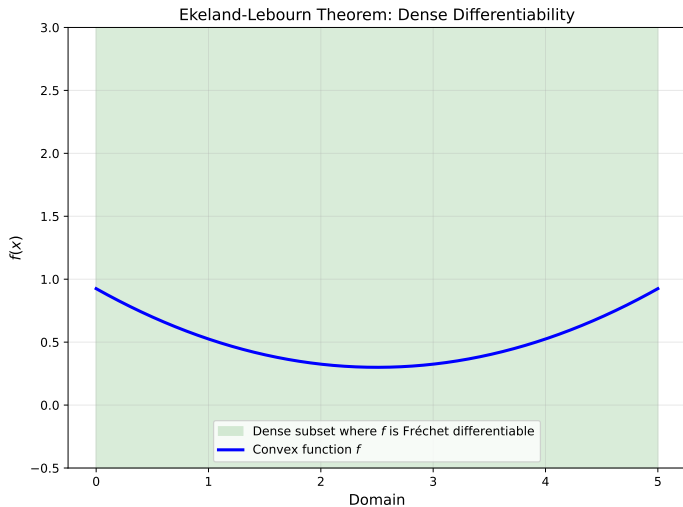
Explanation:

- A G_δ set is a countable intersection of open sets
- “Dense” means the closure equals the whole space
- This is a powerful result: while f may not be differentiable everywhere, it is differentiable “almost everywhere” in a topological sense
- The proof uses:
 - Proposition 18.2 (the set S_ε from the text)

The Ekeland-Lebourn Theorem II

- Continuity properties of convex functions (Theorem 8.38)
- Local boundedness of f

The Ekeland-Lebourn Theorem III



Ekeland-Lebourn: Key Insights

Proof Strategy:

- 1 For $\varepsilon > 0$, define the set:

$$S_\varepsilon = \bigcup_{\eta \in \mathbb{R}_{++}} \left\{ x \in \text{cont } f \mid \sup_{y \in H, \|y\|=1} \frac{f(x + \eta y) + f(x - \eta y) - 2f(x)}{\eta} < \varepsilon \right\}$$

- 2 This set is **open** and **dense** by Proposition 18.2
- 3 The intersection $S = \bigcap_{n \in \mathbb{N} \setminus \{0\}} S_{1/n}$ is a dense G_δ set
- 4 On S , the function f satisfies the Fréchet differentiability condition

Intuition: The oscillations of f become arbitrarily small when restricted to a small enough neighborhood, which forces Fréchet differentiability.

Significance for Convex Analysis:

- Shows that convexity + continuity $\Rightarrow$ “almost everywhere” smoothness
- Motivates the study of subdifferentials (more general than gradients)
- Foundation for optimization algorithms that require differentiability

Subdifferential of a Maximum Function I

Problem: How do subdifferentials behave when we take the pointwise maximum of multiple convex functions?

Theorem (Dubovitskii and Milyutin)

Let $(f_i)_{i \in I}$ be a finite family of convex functions from $\mathcal{H}$ to $[-\infty, +\infty]$, and suppose that $x \in \bigcap_{i \in I} \text{dom } f_i$.

Define $f = \max_{i \in I} f_i$ and $I(x) = \{i \in I \mid f_i(x) = f(x)\}$.

Then:

$$\partial f(x) = \text{conv} \bigcup_{i \in I(x)} \partial f_i(x).$$

Key Points:

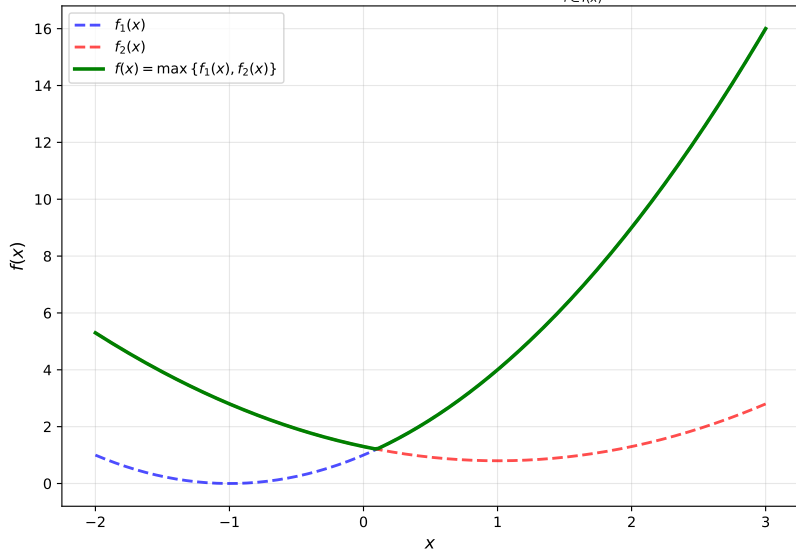
- $I(x)$ is the set of indices where the maximum is achieved
- The subdifferential of the max is the convex hull of the union of subdifferentials at active functions

Subdifferential of a Maximum Function II

- This is the *subdifferential calculus rule for maximum*

Subdifferential of a Maximum Function III

$$\text{Subdifferential of Maximum: } \partial f(x) = \text{conv} \bigcup_{i \in I(x)} \partial f_i(x)$$



Dubovitskii-Milyutin: Proof Outline

Proof uses contradiction:

- 1 Assume strict inclusion: $\partial f(x) \setminus \text{conv} \bigcup_{i \in I(x)} \partial f_i(x) \neq \emptyset$
- 2 Let u be in this difference. By Theorem 3.50 (separation theorem), there exist $y \in \mathcal{H} \setminus \{0\}$ and $\varepsilon > 0$ such that:

$$\langle y | u \rangle \geq \varepsilon + \max_{i \in I(x)} \sup \langle y | \partial f_i(x) \rangle$$

- 3 For $u \in \partial f(x)$, convexity gives:

$$\langle y | u \rangle \geq \max_{i \in I(x)} f'_i(x; y) = f'(x; y)$$

- 4 But $u \notin \text{conv} \bigcup_{i \in I(x)} \partial f_i(x)$ leads to a contradiction

Result: The subdifferential is completely determined by the active functions.

Infimal Convolutions: Overview I

Definition: The *infimal convolution* of $f, g \in \Gamma_0(\mathcal{H})$ is:

$$f \square g(x) = \inf_{y \in \mathcal{H}} \{f(y) + g(x - y)\}$$

Proposition (Differentiability of Infimal Convolution)

Let $f, g \in \Gamma_0(\mathcal{H})$, and let $x, y \in \mathcal{H}$.

Suppose that $x \in \text{dom} \partial(f \square g)$, $f \square g(x) = f(y) + g(x - y)$, and f is Gâteaux differentiable at y . Then:

$$\partial(f \square g)(x) = \{\nabla f(y)\}.$$

Key Insight:

- The infimal convolution simplifies the subdifferential
- When f is differentiable at the minimizer y , the subdifferential becomes a singleton

- This is crucial for variational analysis and optimization

Corollary

If f is real-valued, supercoercive, and Fréchet differentiable on $\mathcal{H}$, then $f \square g$ is Fréchet differentiable on $\mathcal{H}$ with:

$$\nabla(f \square g) = \nabla f.$$

Strict Convexity and Differentiability: Duality

Central Question: What is the relationship between differentiability of f and strict convexity of f^* ?

Proposition (Differentiability implies strict convexity of dual)

Let $f \in \Gamma_0(\mathcal{H})$ be such that f^* is strictly convex on every nonempty convex subset of $\text{dom } \partial f^*$.
Then f is Gâteaux differentiable on $\text{int dom } f$.

Proposition (Strict convexity implies differentiability of dual)

Let $f \in \Gamma_0(\mathcal{H})$ be Gâteaux differentiable on $\text{int dom } f$.
Then f^* is strictly convex on every nonempty convex subset of $\nabla f(\text{int dom } f)$.

Corollary (Complete Characterization)

Suppose $\text{dom } \partial f = \text{int dom } f$ and $\text{dom } \partial f^ = \text{int dom } f^*$.
Then f is Gâteaux differentiable on $\text{int dom } f$ if and only if f^* is strictly convex on $\text{int dom } f^*$.*

Stronger Differentiability Concepts I

Beyond Fréchet differentiability: We study gradient continuity and uniform growth.

Theorem (Characterization of Lipschitz Continuous Gradients)

Let $f : \mathcal{H} \rightarrow \mathbb{R}$ be Fréchet differentiable, and let $\phi : \mathbb{R} \rightarrow \mathbb{R}_+$ be an even convex function that vanishes only at 0.

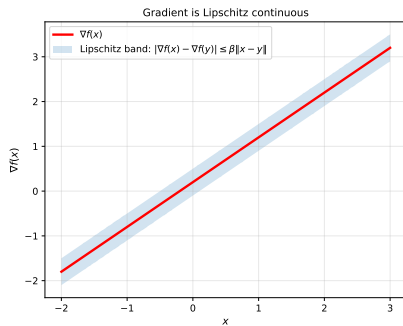
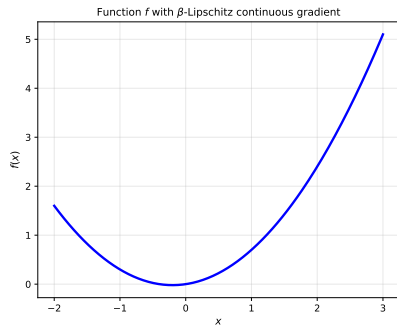
Define $\psi(s) = \frac{\phi(s)}{|s|}$ for $s \neq 0$, $\psi(0) = 0$, and $\theta(s) = \int_0^1 \frac{\phi(st)}{t} dt$.

Then all of the following are equivalent:

- (i) $(\forall x, y \in \mathcal{H})(\|\nabla f(x) - \nabla f(y)\| \leq \psi(\|x - y\|))$
- (ii) $(\forall x, y \in \mathcal{H})(\langle x - y | \nabla f(x) - \nabla f(y) \rangle \leq \phi(\|x - y\|))$
- (iii) $(\forall x, y \in \mathcal{H})(f(y) \leq f(x) + \langle y - x | \nabla f(x) \rangle + \theta(\|x - y\|))$

And (i) $\Rightarrow$ (ii) $\Rightarrow$ (iii) $\Rightarrow$ (iv) $\Rightarrow$ (v) $\Rightarrow$ (vi).

Stronger Differentiability Concepts II



Lipschitz Continuous Gradients: Important Cases

Special Case: Hölder Continuous Gradients

Corollary (Hölder Continuous Gradients)

Let $f : \mathcal{H} \rightarrow \mathbb{R}$ be Fréchet differentiable, $\beta \in \mathbb{R}_{++}$, and $p \in [0, 1]$. Consider the following statements:

- (i) $(\forall x, y \in \mathcal{H})(\|\nabla f(x) - \nabla f(y)\| \leq \beta \|x - y\|^p)$
- (ii) $(\forall x, y \in \mathcal{H})(\langle x - y | \nabla f(x) - \nabla f(y) \rangle \leq \beta \|x - y\|^{p+1})$
- (iii) $(\forall x, y \in \mathcal{H})(f(y) \leq f(x) + \langle y - x | \nabla f(x) \rangle + \beta(p+1)^{-1} \|x - y\|^{p+1})$

Then $(i) \Rightarrow (ii) \Rightarrow (iii) \Rightarrow (iv) \Rightarrow (v) \Rightarrow (vi)$.

Most Important Special Case: β -Smooth Functions

When $p = 1$, we get the classical notion of β -Lipschitz continuous gradients:

$$\|\nabla f(x) - \nabla f(y)\| \leq \beta \|x - y\|$$

This is also called β -smoothness or the gradient being β -Lipschitz continuous.

Baillon-Haddad Theorem

Corollary (Baillon-Haddad)

Let $f : \mathcal{H} \rightarrow \mathbb{R}$ be a Fréchet differentiable convex function and $\beta \in \mathbb{R}_{++}$. Then ∇f is β -Lipschitz continuous if and only if ∇f is $(1/\beta)$ -cocoercive. In particular, ∇f is nonexpansive if and only if ∇f is firmly nonexpansive.

Cocoercivity Definition: An operator A is α -cocoercive if:

$$\langle x - y | Ax - Ay \rangle \geq \alpha \|Ax - Ay\|^2$$

Significance:

- Links differentiability properties with operator monotonicity
- $(1/\beta)$ -cocoercivity is weaker than β -Lipschitz continuity
- Crucial for monotone operator theory and fixed point algorithms

Moreau Envelope and Regularization

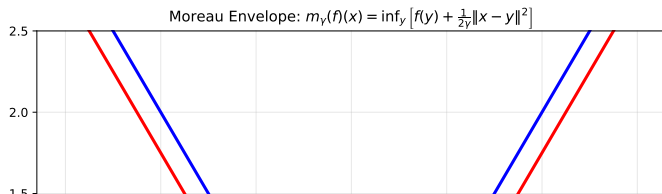
Corollary (Moreau Envelope)

Let $f \in \Gamma_0(\mathcal{H})$, $\beta \in \mathbb{R}_{++}$, and $q = (1/2)\|\cdot\|^2$. Then f is Fréchet differentiable on $\mathcal{H}$ with a β -Lipschitz continuous gradient if and only if f is the Moreau envelope of parameter $1/\beta$ of a function in $\Gamma_0(\mathcal{H})$:

$$f = \left(f^* - \frac{1}{\beta}q\right)^* \square \beta q.$$

Moreau Envelope:

$$m_\gamma(f)(x) := \inf_y \left\{ f(y) + \frac{1}{2\gamma} \|x - y\|^2 \right\}$$



Proximal Operators and Averaged Proximal Operators

Corollary (Averaged Proximal Operators)

Let $(f_i)_{i \in I}$ be a finite family of functions in $\Gamma_0(\mathcal{H})$, let $(\alpha_i)_{i \in I}$ be a finite family of real numbers in $[0, 1]$ such that $\sum_{i \in I} \alpha_i = 1$, and set $q = (1/2)\|\cdot\|^2$.

Then $\sum_{i \in I} \alpha_i \text{Prox}_{f_i}$ is the proximity operator of a function $h \in \Gamma_0(\mathcal{H})$:

$$\sum_{i \in I} \alpha_i \text{Prox}_{f_i} = \text{Prox}_h, \quad \text{where } h = \left(\sum_{i \in I} \alpha_i (f_i^* \square q) \right)^* - q.$$

Key Application:

- Shows how to average proximal operators
- Important for parallel optimization algorithms
- Special case: $\alpha_1 = \alpha_2 = 1/2$ gives the proximal average

Distance Function to a Convex Set I

Definition: For a nonempty closed convex set $C \subseteq \mathcal{H}$:

$$d_C(x) = \inf_{z \in C} \|x - z\|$$

Question: When is d_C differentiable, and what is its subdifferential?

Proposition (Differentiability Characterization)

Let C be a nonempty closed convex subset of $\mathcal{H}$ and let $x \in C$.

Then the following are equivalent:

- (i) d_C is Gâteaux differentiable at x and $\nabla d_C(x) = 0$.
- (ii) $x \notin \text{spts } C$, i.e., $(\forall u \in \mathcal{H} \setminus \{0\}) \sigma_C(u) > \langle x | u \rangle$.
- (iii) $T_C x = \mathcal{H}$, i.e., $\overline{\text{cone}(C - x)} = \mathcal{H}$ (tangent cone is the whole space).

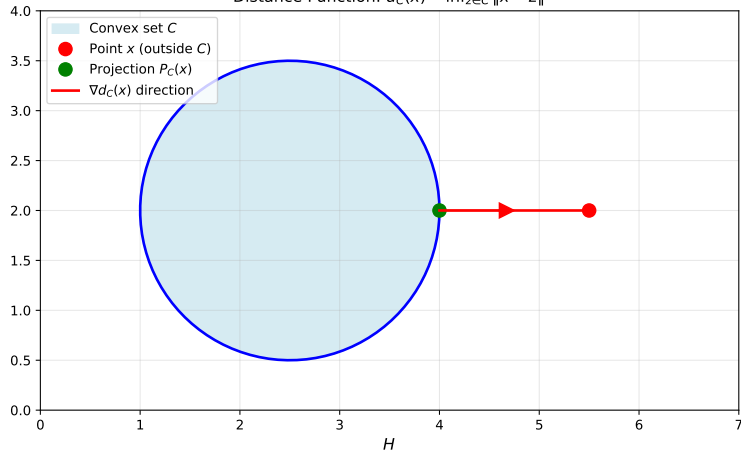
Interpretation:

Distance Function to a Convex Set II

- Points on the interior of C have zero gradient of the distance function
- Points not in the support of C satisfy stronger properties
- Tangent cone condition (iii) ensures full-dimensionality locally

Distance Function to a Convex Set III

Distance Function: $d_C(x) = \inf_{z \in C} \|x - z\|$



Distance Function: Cases at the Boundary

Proposition (Complete Characterization for All Points)

Let C be a nonempty closed convex subset of $\mathcal{H}$ and $x \in \mathcal{H}$.

Then exactly one of the following holds:

- (i) Suppose that $x \in \text{int } C$. Then d_C is Fréchet differentiable at x with $\nabla d_C(x) = 0$.
- (ii) Suppose that $x \in \text{bdry } C$. Then:
 - (a) If $x \notin \text{spts } C$: d_C is not Fréchet differentiable but is Gâteaux differentiable with $\nabla d_C(x) = 0$.
 - (b) If $x \in \text{spts } C$: d_C is not Gâteaux differentiable and $\partial d_C(x) = N_C x \cap B(0; 1)$.
- (iii) Suppose that $x \notin C$. Then d_C is Fréchet differentiable at x with $\nabla d_C(x) = \frac{x - P_C x}{d_C(x)}$.

Key Point: The distance function has three distinct behaviors depending on whether x is interior, boundary, or exterior to C .

Example: Convex Quadratic Function

Problem: Analyze differentiability of $f(x) = \frac{1}{2}\|x\|^2$.

```
import numpy as np
import matplotlib.pyplot as plt

# Quadratic function:  $f(x) = 0.5 * \|x\|^2$ 
def f(x):
    return 0.5 * np.linalg.norm(x)**2

# Gradient (Frchet differentiable):  $\text{grad } f(x) = x$ 
def grad_f(x):
    return x

# Subdifferential at  $x$ : singleton  $\{x\}$ 
def subdiff_f(x):
    return {np.array(x)}

# Verify Lipschitz continuity of gradient
x1 = np.array([1.0, 2.0, 3.0])
x2 = np.array([1.1, 2.1, 3.1])

grad_x1 = grad_f(x1)
grad_x2 = grad_f(x2)

#  $\| \text{grad } f(x1) - \text{grad } f(x2) \| = \| x1 - x2 \|$ 
diff_grad = np.linalg.norm(grad_x1 - grad_x2)
diff_x = np.linalg.norm(x1 - x2)
```

Example: Subdifferential of Maximum

Problem: Compute $\partial f(x)$ where $f(x) = \max\{f_1(x), f_2(x)\}$ with:

$$f_1(x) = \frac{1}{2}(x+1)^2, \quad f_2(x) = \frac{1}{2}(x-1)^2$$

```
import numpy as np

def f1(x):
    return 0.5 * (x + 1)**2

def f2(x):
    return 0.5 * (x - 1)**2

def grad_f1(x):
    return x + 1

def grad_f2(x):
    return x - 1

def subdiff_max(x):
    """Subdifferential of max{f1, f2}"""
    f1_val = f1(x)
    f2_val = f2(x)
    I_x = []

    if np.isclose(f1_val, f2_val):
```

Example: Distance Function to a Disk

Problem: Analyze the distance function to the closed disk:

$$C = \{x \in \mathbb{R}^2 : \|x\| \leq r\}$$

```
import numpy as np

def distance_to_disk(x, r=1.0):
    """Distance from x to closed disk of radius r"""
    x_norm = np.linalg.norm(x)
    return max(0, x_norm - r)

def grad_distance_outside(x, r=1.0):
    """Gradient of distance function when x is outside disk"""
    x_norm = np.linalg.norm(x)
    if x_norm <= r:
        return None # Not outside
    return x / x_norm # Unit vector from origin

# Test points
x_inside = np.array([0.2, 0.3]) # Inside disk
x_boundary = np.array([1.0, 0.0]) # On boundary
x_outside = np.array([2.0, 0.0]) # Outside disk

print(f"x_inside: d_C(x) = {distance_to_disk(x_inside):.4f}")
print(f"x_boundary: d_C(x) = {distance_to_disk(x_boundary):.4f}")
print(f"x_outside: d_C(x) = {distance_to_disk(x_outside):.4f}")
```

Key Takeaways

Main Results of Chapter 18:

- (i) **Ekeland-Lebourn:** Convex + continuous $\Rightarrow$ dense set of Fréchet differentiability
- (ii) **Maximum Rule:** $\partial \max_i f_i = \text{conv} \bigcup_i \partial f_i$ (for active indices)
- (iii) **Infimal Convolution:** Differentiability of f at optimizer $\Rightarrow$ singleton subdifferential of $f \square g$
- (iv) **Duality:** Differentiability of $f \Leftrightarrow$ strict convexity of f^*
- (v) **Lipschitz Gradients:** Multiple equivalent characterizations of smooth/Lipschitz gradient property
- (vi) **Baillon-Haddad:** β -Lipschitz gradient $\Leftrightarrow (1/\beta)$ -cocoercivity
- (vii) **Distance Function:** Complete characterization of differentiability based on position relative to set

Applications in Optimization

Practical Uses of Chapter 18 Results:

- **Proximal Gradient Methods:**

- Moreau envelope provides smooth approximation
- Lipschitz gradient characterization ensures convergence rates

- **Proximal Point Algorithm:**

- Uses averaged proximal operators (Corollary 18.20)
- Guaranteed convergence via monotone operator theory

- **Constraint Optimization:**

- Distance function characterization handles constraints
- Subdifferential of max handles multiple constraints

- **Duality Theory:**

- Dual variables recovered from primal solutions
- Strict convexity of dual $\Leftrightarrow$ smoothness of primal

- **Numerical Stability:**

- Smooth functions allow gradient-based methods
- Lipschitz constants control error bounds

Further Reading and References

Extensions of these results:

- **Higher-order differentiability:** Gâteaux vs. Fréchet second derivatives
- **Smooth approximations:** Moreau envelope generalizations, Huber loss
- **Variational analysis:** Coderivatives, graphical convergence
- **Monotone operator theory:** Connection between subdifferentials and monotone operators (Chapter 20-21)
- **Machine learning:** Lipschitz gradient bounds in neural networks, generalization theory
- **Stochastic optimization:** Large-scale versions of proximal methods

Key Reference:

Convex Analysis and Monotone Operator Theory in Hilbert Spaces
Heinz H. Bauschke and Patrick L. Combettes (2017)
Springer-Verlag

Appendix: Descent Lemma I

Theorem 18.15(iii): The Descent Lemma

For a Fréchet differentiable convex function f with β -Lipschitz continuous gradient:

$$f(y) \leq f(x) + \langle y - x | \nabla f(x) \rangle + \frac{\beta}{2} \|y - x\|^2$$

Proof:

Let $x, y \in \mathcal{H}$. By the mean value theorem applied to the function $\phi_z(t) = f(x + t(z - x))$:

$$f(y) - f(x) = \int_0^1 \langle y - x | \nabla f(x + t(y - x)) \rangle dt \tag{1}$$

$$= \langle y - x | \nabla f(x) \rangle + \int_0^1 \langle y - x | \nabla f(x + t(y - x)) - \nabla f(x) \rangle dt \tag{2}$$

Appendix: Descent Lemma II

Using the Lipschitz property of ∇f :

$$\|\nabla f(x + t(y - x)) - \nabla f(x)\| \leq \beta \cdot t \|y - x\|$$

By Cauchy-Schwarz:

$$\langle y - x | \nabla f(x + t(y - x)) - \nabla f(x) \rangle \leq \beta t \|y - x\|^2$$

Therefore:

$$\int_0^1 \langle y - x | \nabla f(x + t(y - x)) - \nabla f(x) \rangle dt \leq \int_0^1 \beta t \|y - x\|^2 dt = \frac{\beta}{2} \|y - x\|^2$$

Appendix: Baillon-Haddad Proof Sketch

Theorem 18.17: Baillon-Haddad

We show: β -Lipschitz gradient $\Leftrightarrow (1/\beta)$ -cocoercivity

Forward direction ($\Rightarrow$):

Assume $\|\nabla f(x) - \nabla f(y)\| \leq \beta\|x - y\|$.

By Proposition 17.35 (Fenchel-Young), for any $u \in \partial f(x)$:

$$f^*(u) \geq \langle u|x \rangle - f(x)$$

For a Fréchet differentiable f , we have $\partial f(x) = \{\nabla f(x)\}$, so:

$$f^*(\nabla f(x)) = \langle \nabla f(x)|x \rangle - f(x)$$

The $(1/\beta)$ -cocoercivity follows from the descent lemma and properties of the convex conjugate.

Backward direction ($\Leftarrow$):

Assume $(1/\beta)$ -cocoercivity: $\langle x - y | \nabla f(x) - \nabla f(y) \rangle \geq \frac{1}{\beta} \|\nabla f(x) - \nabla f(y)\|^2$

This implies the Lipschitz continuity via the relationship between monotonicity and Lipschitz properties.